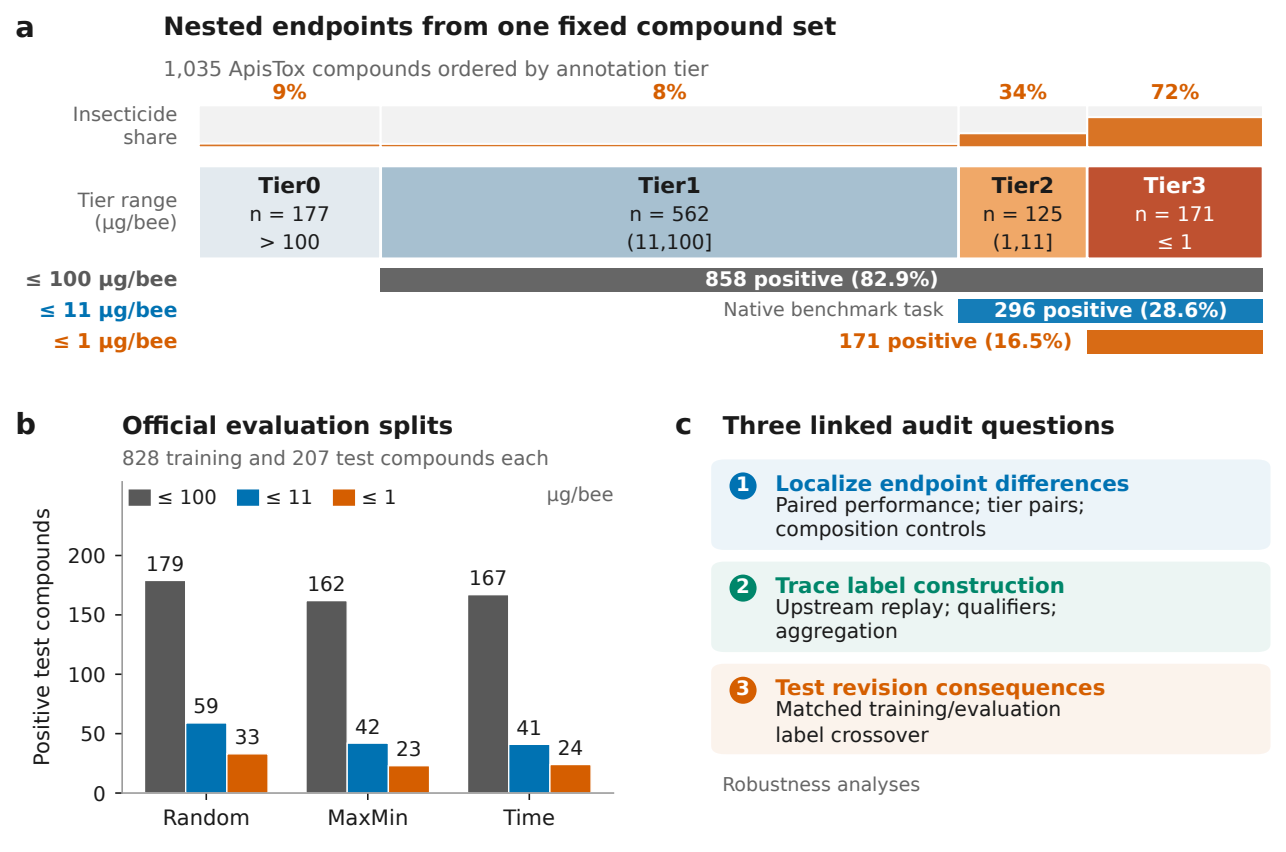
a
Nested endpoints from one fixed compound set
1,035 ApisTox compounds ordered by annotation tier
9%
8%
34%
72%
Insecticide share
Tier range (µg/bee)
Tier0
n = 177
> 100
Tier1
n = 562
(11,100]
Tier2
n = 125
(1,11]
Tier3
n = 171
≤ 1
≤ 100 µg/bee
858 positive (82.9%)
≤ 11 µg/bee
Native benchmark task
296 positive (28.6%)
≤ 1 µg/bee
171 positive (16.5%)
b
Official evaluation splits
828 training and 207 test compounds each
≤ 100
≤ 11
≤ 1
µg/bee
Positive test compounds
0
50
100
150
200
179
59
33
162
42
23
167
41
24
Random
MaxMin
Time
c
Three linked audit questions
1
Localize endpoint differences
Paired performance; tier pairs; composition controls
2
Trace label construction
Upstream replay; qualifiers; aggregation
3
Test revision consequences
Matched training/evaluation label crossover
Robustness analyses